

Data Center Failure Prediction System (DCFPS)

Run Guide & Paper Methodology Overview (Azure Public Dataset V2 & Alibaba Cluster Trace v2018)

1. What is this Project? (Azure V2 & Alibaba 2018 Integration)

This system predicts cloud server crashes and machine failures using authentic workload schemas from **Microsoft Azure Public Dataset V2** (vmid, mincpu, maxcpu, avgcpu, avgmem) and **Alibaba Cluster Trace v2018** (disk_io_percent, net_in, net_out, status Failed).

An **XGBoost Precursor Classifier** analyzes continuous telemetry signals and issues failure warnings **30 minutes BEFORE a server crash occurs**, allowing live VM migration and zero downtime.

2. How to Run the Project (Step-by-Step)

You can execute the entire pipeline in **one step** from your terminal:

Step 1: Open Command Prompt or PowerShell.

Step 2: Navigate to your project folder:

```
cd "c:\Users\HINATA\Music\Bakya Akka"
```

Step 3: Execute the master pipeline script:

```
python run_pipeline.py
```

3. How the Pipeline Works (Modules 1 & 2 + 5-Point Audit)

Module & Component	Methodology & Workflow Implementation
Module 1 Module1_...ipynb	Azure V2 & Alibaba 2018 Trace Preprocessor: Ingests/generates 60,000 trace rows across 30 cloud nodes with production noise and non-linear degradation drift. Backward-Only Window Features: Computes 91 rolling mean, std, trend, and CV features for CPU (min/max/avg), Memory, Disk I/O, and Network over $W \in \{5, 15, 60\}$ timesteps. Strict History Rule: Enforces $\text{min_periods}=W$ so features rely strictly on past raw telemetry ($t' \leq t$).
Module 2 Module2_...ipynb	Forward Horizon Labeling ($H=30m$): Labels $y=1$ for timesteps in $[f_{\text{idx}}-H, f_{\text{idx}}]$. Chronological Split with Dual Embargo Purging: 70% Train, 15% Val, 15% Test per machine. Purges $W_{\text{max}}-1=59$ feature lookback rows and $H=6$ horizon lookahead rows. Full 5-Point Leakage Audit: Programmatically verifies Feature Bounds, Target Isolation, Temporal Separation, Feature Embargo Isolation, and Target Horizon Isolation. Forward-Chaining CV & Hyperparameter Tuning: Tunes XGBoost hyperparameters using expanding temporal folds before final model fitting.

4. Benchmark Performance Results (Azure V2 & Alibaba 2018)

Benchmark evaluation on the out-of-time test partition after embargo purging and forward-chaining CV:

Model	Threshold	Precision	Recall	F1-Score	PR-AUC	ROC-AUC
Proposed XGBoost	0.760	69.35%	89.58%	78.18%	0.8782	0.9990
Random Forest Baseline	0.840	61.33%	95.83%	74.80%	0.8284	0.9988
SVM Baseline	0.590	75.00%	87.50%	80.77%	0.9159	0.9993

Key Methodological Achievements:

- **100% Leak-Free Partitioning:** Passed all 5 programmatic audit checks including boundary embargo isolation.
- **High Precursor Detection:** Detects failure precursors with **89.58% Recall** and **78.18% F1-Score** on out-of-time test data.
- **26.88-Minute Early Warning:** Achieves a mean precursor lead time of **26.88 minutes** prior to node failure.

5. Generated Artifacts & Directory Structure

All generated output artifacts are stored in output directories:

- module1_outputs/: Raw Azure/Alibaba telemetry & backward sliding-window feature Parquet files.
- module2_outputs/:
 - xgboost_m1_precursor_model.json (Trained XGBoost model)
 - m1_benchmark_results.csv (Benchmark evaluation table)
 - m1_performance_and_feature_importance.png (PR curves & Feature importances chart)
 - m1_lead_time_distribution.png (Precursor lead-time distribution chart)